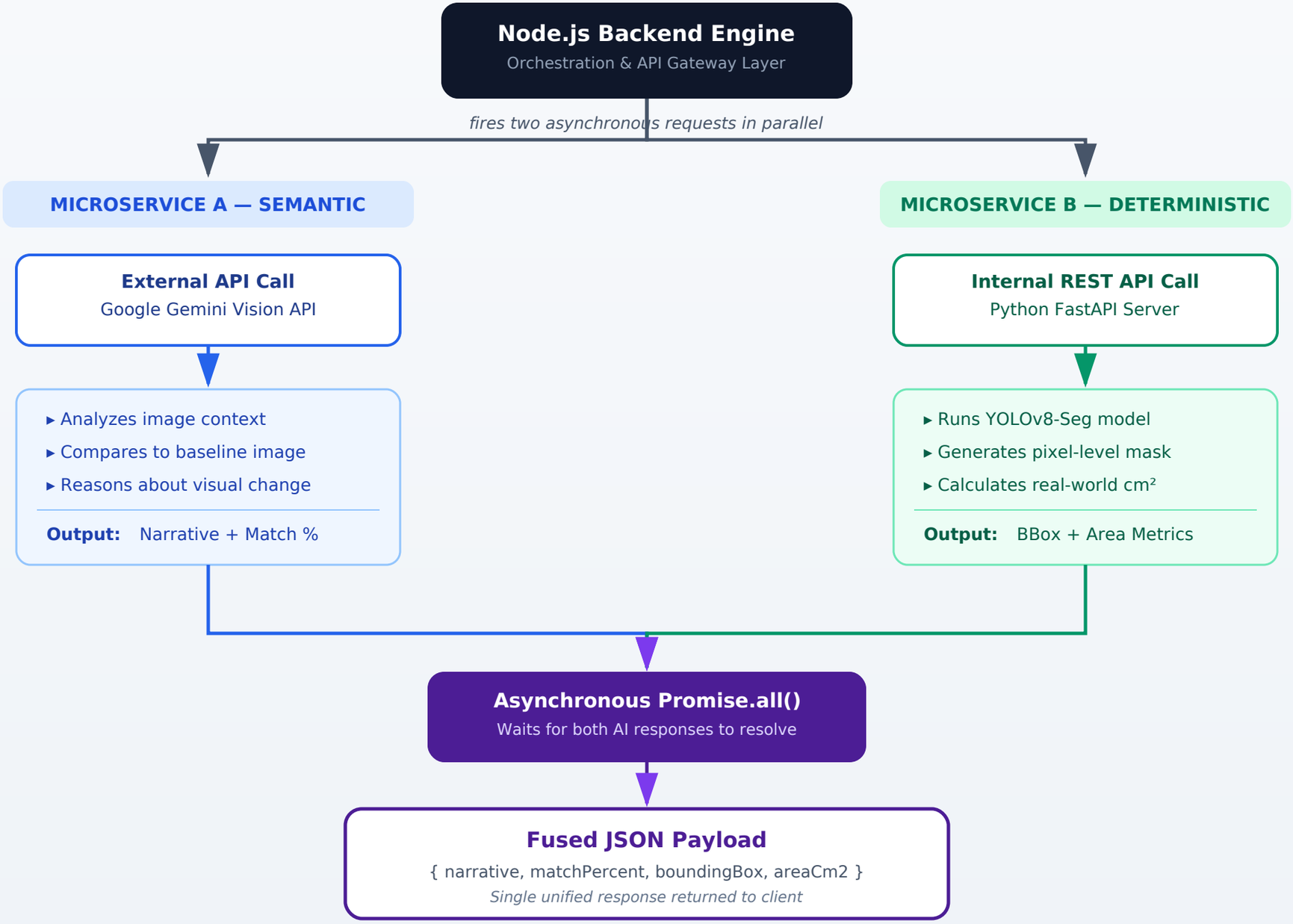
# Dual-Pipeline AI Fusion Architecture

Parallel Semantic + Deterministic Analysis with Asynchronous Result Fusion




**Legend**

- Semantic reasoning path (Gemini Vision — qualitative, context-aware)
- Deterministic CV path (YOLOv8-Seg — quantitative, pixel-precise)